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# System Requirements Specification

for

## AStra: The Multiverse for Creators

Creator Relationship Commerce and Collaboration Platform  
Version 0.3

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## Revision History

| Version | Date        | Author          | Comment                                                                                                                                |
|---------|-------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 0.1     | 09/25/2026  | Saloni and Arya | Initial draft using the supplied IEEE SRS format                                                                                       |
| 0.2     | 09/25/2026  | Saloni and Arya | Added creator-only collaboration feed, profile filters, and response limits                                                            |
| 0.3     | 09/26/2026  | Saloni and Arya | Added role-based workflows, creator commerce, connected apps, shared loyalty points, checkout confirmation, and administrator controls |
| 1.0     | To be added | Team            | Release version                                                                                                                        |

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# 1 Introduction

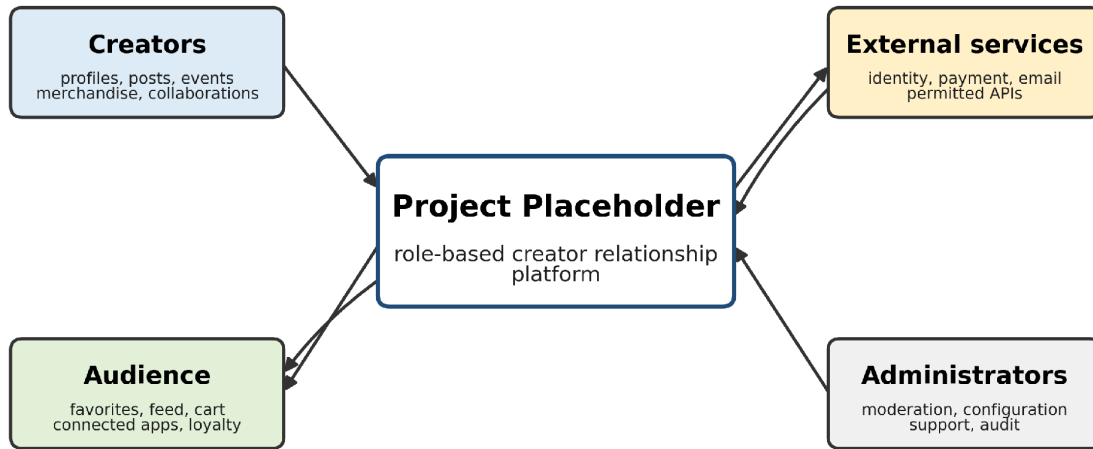
Independent creators build audiences across social media, streaming services, email lists, stores, ticketing products, and private communities. Each service holds only one part of the creator and audience relationship. Creators can struggle to reach supporters consistently, publish and sell work from one place, recognize meaningful participation, and find trustworthy collaborators. Audience members can miss important announcements, manage purchases across separate tools, and receive little recognition for the different ways they support creative work.

This project specifies a web platform that helps creators manage profiles, publish posts, events, and merchandise, sell eligible items through a cart and checkout flow, build permission-based audience relationships, configure transparent loyalty benefits, and collaborate across disciplines. Audience members receive a combined feed, favorites, connected-app controls, cart and order history, and a clear loyalty balance. Administrators manage verification, platform rules, moderation, disputes, and audit records. The specification is a working product agreement for Saloni Belliappa Bolakaranda as Product Manager and Arya Jay Wadhvani as Technical Lead. Survey results, interviews, product naming, payment-provider selection, and external platform permissions remain to be confirmed.

## 1.1 Project Scope

The first version is a responsive web application with separate creator, audience, and administrator dashboards. Creators can manage a profile, publish posts, events, and merchandise, review audience and order activity, configure loyalty rules within platform limits, and create, join, or manage collaborations. Audience members can favorite creators, use a combined community feed, connect eligible apps, maintain a cart, complete checkout, review orders, earn loyalty points, and apply permitted discounts. Administrators can manage verification, moderation, loyalty settings, commerce exceptions, and audit records. The platform complements existing social, event, payment, email, and streaming services rather than replacing them.

## Creator relationship commerce and collaboration ecosystem



**Figure 1-1: Creator relationship commerce and collaboration ecosystem**

The hackathon prototype includes creator and audience accounts, role-based dashboards, creator profiles, posts, events, merchandise listings, favorite creators, a community feed, connected-app permission controls, a cart and simulated or sandbox checkout, order confirmation email, one directly verifiable or simulated audience activity, a private loyalty ledger, one discount conversion rule, cross-genre creator discovery, collaboration creation and participation, basic notifications and analytics, and administrator safety controls.

The prototype excludes a new streaming service, a fully liquid marketplace, automated legal or copyright enforcement, direct creator payouts outside the selected payment provider, cash redemption or peer-to-peer transfer of loyalty points, guaranteed access to every external platform, advanced tax or accounting functions, complex fulfillment, and native mobile applications. Identity, listening, viewing, and payment data may use sandbox or simulated flows until provider permission and reliable verification are confirmed.

## 1.2 Acronyms and Definitions

Table 1-1 lists acronyms used in this document. Table 1-2 defines important product terms so that product, engineering, mentors, judges, and research participants use the same meaning.

**Table 1-1: Acronym list**

| Acronym | Definition                          |
|---------|-------------------------------------|
| API     | Application Programming Interface   |
| MVP     | Minimum Viable Product              |
| PII     | Personally Identifiable Information |
| SRS     | Software Requirements Specification |

| Acronym | Definition                         |
|---------|------------------------------------|
| UI      | User Interface                     |
| HTTPS   | Hypertext Transfer Protocol Secure |
| IP      | Intellectual Property              |

**Table 1-2: Glossary of terms**

| Term                           | Definition                                                                                                                                                                            |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Creator                        | A person or team producing creative or educational work, including musicians, video creators, writers, artists, dancers, photographers, streamers, and educators.                     |
| Audience member                | A person who follows, watches, listens to, buys from, attends, or otherwise supports a creator. The workflow diagrams use the shorter label Audience.                                 |
| Owned audience                 | People who have directly permitted a creator to contact or understand them, rather than being reachable only through another platform's algorithm.                                    |
| Activity                       | An action a creator chooses to recognize, such as an RSVP, event check-in, purchase code, poll, referral, or mailing-list signup.                                                     |
| Reward                         | A creator-approved benefit such as recognition, early access, special content, a discount, or an experience.                                                                          |
| Loyalty ledger                 | A private history that explains which activities were recognized and how an audience member's status or benefit was calculated.                                                       |
| Creator collaboration          | A project involving two or more creators, often from different creative disciplines.                                                                                                  |
| Collaboration brief            | A plain-language record of project roles, deliverables, dates, credit, intended use, approvals, and any proposed payment or revenue split. It is not a legal contract.                |
| Collaboration opportunity      | A creator-only post seeking another creator for a specific project. It states the work, desired profile, arrangement, response limit, and timing.                                     |
| Opportunity response           | A single expression of interest linked to one collaboration opportunity. It contains a short message and the responding creator's profile or portfolio, not a formal job application. |
| Required and preferred filters | Required filters determine who may respond. Preferred filters highlight stronger matches without preventing other eligible creators from responding.                                  |
| Cart                           | A temporary collection of event tickets or merchandise that an audience member intends to review and purchase.                                                                        |
| Order                          | A recorded purchase created after successful checkout, with status, items, totals, loyalty discount, and confirmation history.                                                        |
| Connected app                  | An external account or service that a user explicitly authorizes for a stated purpose and can disconnect later.                                                                       |
| Loyalty point                  | A platform-wide, non-cash unit recorded in a private ledger and redeemable only under an active discount or benefit rule.                                                             |
| Rights metadata                | Creator-supplied ownership, credit, license, or intended-use information attached to content. It does not establish or enforce legal rights.                                          |

## 2 Overall Description

The product is a creator relationship, commerce, loyalty, and collaboration platform. It sits between creators, audience members, administrators, and selected identity, payment, email, social, event, and streaming services. Role-specific dashboards expose only the functions allowed for each account. The system stores only the information needed for visible features and remains useful when an optional external account is not connected.

### 2.1 System Context

Creators provide profiles, posts, events, merchandise, rights metadata, loyalty rules, skills, and collaboration opportunities. Audience members provide favorites, cart and checkout choices, connected-app permissions, eligible activity, redemptions, and feedback. External services may verify identity, process a sandbox or live payment, send email, provide authorized activity, or receive an allowed link. Administrators manage verification, platform loyalty settings, reports, commerce or reward disputes, account actions, and audit records.

#### System context and information exchange

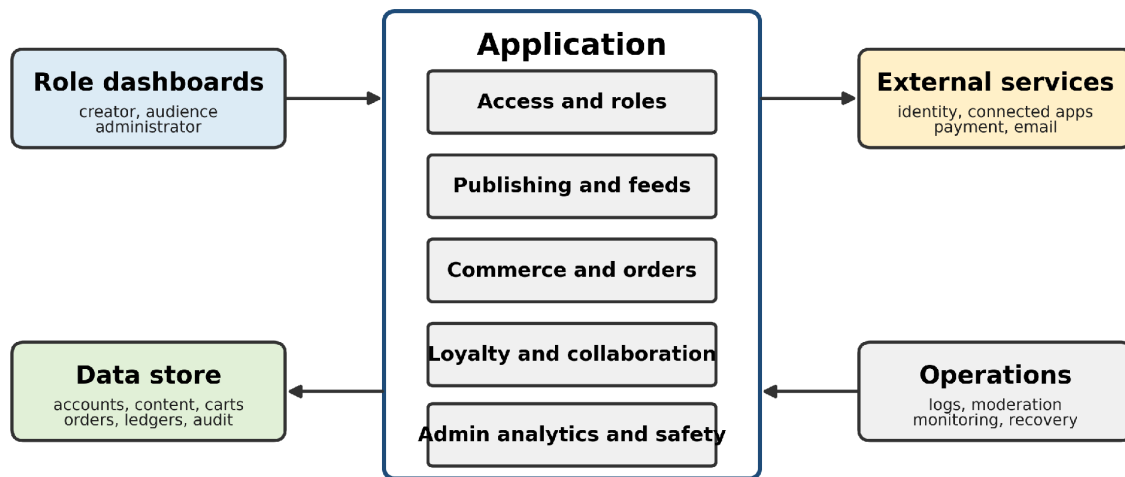


Figure 2-1: System context and information exchange

Table 2-1: Stakeholders and needs

| Stakeholder      | Primary need                                                                                                       |
|------------------|--------------------------------------------------------------------------------------------------------------------|
| Creators         | Reach audiences, publish and sell work, understand engagement, collaborate, and protect trust.                     |
| Audience members | Find reliable updates, manage purchases, earn clear benefits, and control personal information and connected apps. |

| Stakeholder                 | Primary need                                                                                                    |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------|
| Creator collaborators       | Find complementary people, agree on the work, and reduce confusion about credit, use, and money.                |
| Product Manager             | Validate the problem, define scope and acceptance criteria, and present evidence.                               |
| Technical Lead              | Build a stable prototype and state technical or integration limits honestly.                                    |
| Administrators              | Handle verification, reports, bad actors, suspicious rewards, order exceptions, loyalty settings, and disputes. |
| External platform providers | Ensure identity, payment, email, and connected-app use follows current technical and policy rules.              |

## 2.2 System Functions

The system supports the following functions:

- Create and manage creator and audience accounts, provision administrator accounts, verify creator identity where required, and route each role to an appropriate dashboard.
- Let creators manage profiles and rights metadata, publish posts, events, and merchandise, and create or join collaboration opportunities.
- Let audience members search for creators, save favorites, view a community feed, control connected apps, and review a private cart and order history.
- Let creators define eligible audience activities and rewards while administrators configure platform-wide loyalty point and discount limits.
- Maintain a private loyalty ledger with source, date, points, discount use, and status for recognized activity.
- Help creators contact one another directly or use a creators-only opportunity feed to discover complementary skills, respond within controlled limits, and exchange a structured project brief.
- Support carts, checkout, payment status, order records, email confirmations, basic analytics, reports, moderation, and auditable administrator actions.
- Support explicit connected-app permissions and selected identity, payment, and email services while clearly labeling sandbox or simulated data in the prototype.

**Table 2-2: User groups**

| User group                  | Typical situation                                 | Primary need                                                                   |
|-----------------------------|---------------------------------------------------|--------------------------------------------------------------------------------|
| Emerging creator            | Fewer than 5,000 followers and several free tools | Affordable discovery, collaboration, and simple promotion                      |
| Growing creator             | 5,000 to 100,000 followers and some revenue       | Identify likely supporters, improve conversion, and build repeat relationships |
| Established creator or team | Regular launches, events, merchandise, or staff   | Segmentation, team access, automation, and reliable reporting                  |

| User group               | Typical situation                                                                 | Primary need                                                                |
|--------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Casual audience member   | Follows several creators but rarely pays                                          | Important updates and low-effort participation                              |
| Active supporter         | Attends, buys, shares, participates in communities, and may connect eligible apps | Recognition, early access, and meaningful experiences                       |
| Cross-genre collaborator | Needs a skill or audience different from their own                                | Relevant discovery, trust signals, clear scope, and shared credit           |
| Administrator            | Supports a pilot with creators, audience members, rewards, and orders             | Verification, moderation, configuration, dispute handling, and auditability |

## 2.3 Role Based Workflows

The updated workflow uses one entry point and routes authenticated users to a creator, audience, or administrator dashboard. Public registration supports creator, audience, or combined creator-audience access; administrator privileges are assigned through a managed process. The diagrams below define the navigation and state changes that the detailed requirements must support.

### Role entry and dashboard routing

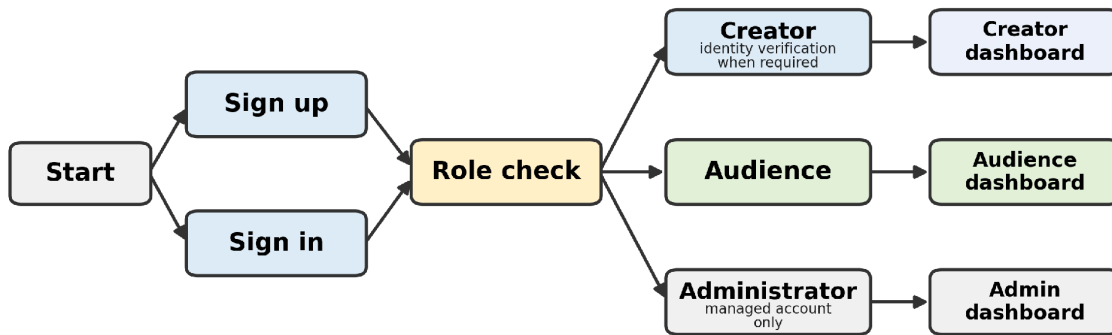
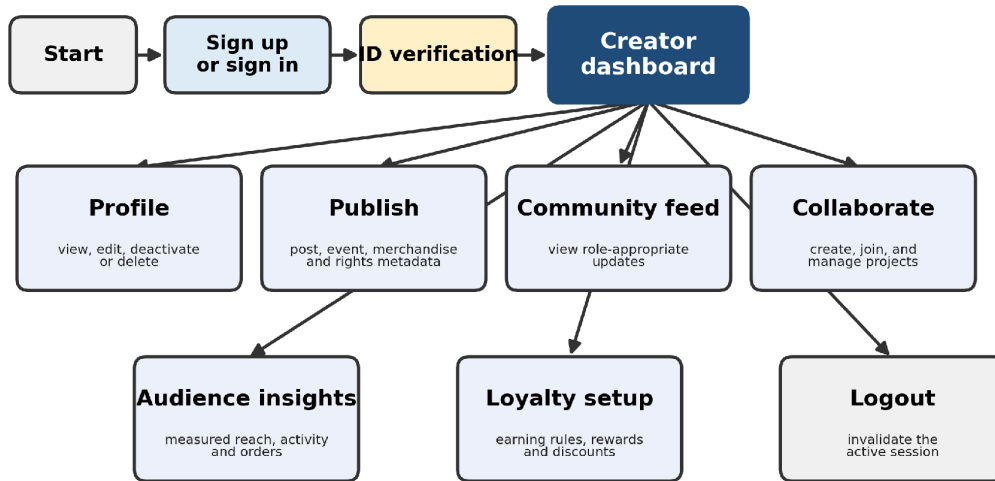


Figure 2-2: Role entry and dashboard routing

Creator verification is required before high-risk publishing, loyalty configuration, or payment-related actions when the pilot policy requires it. A user with both creator and audience access can switch dashboards without creating a duplicate account.

## Creator dashboard workflow

Creator actions from account access through publishing collaboration and account exit

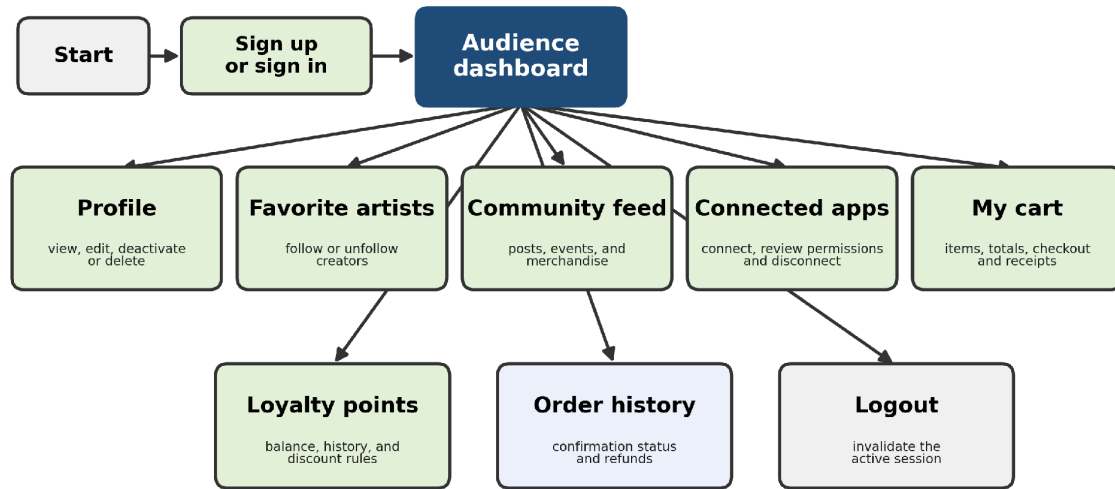


**Figure 2-3: Creator dashboard workflow**

The creator dashboard is the operational home for profile management, publishing, community activity, collaboration, audience insights, loyalty setup, and logout. Publishing separates posts, events, and merchandise so each item type can enforce its own required fields and status rules. Rights metadata records creator-provided ownership and intended-use information but does not replace legal review or enforcement.

## Audience dashboard workflow

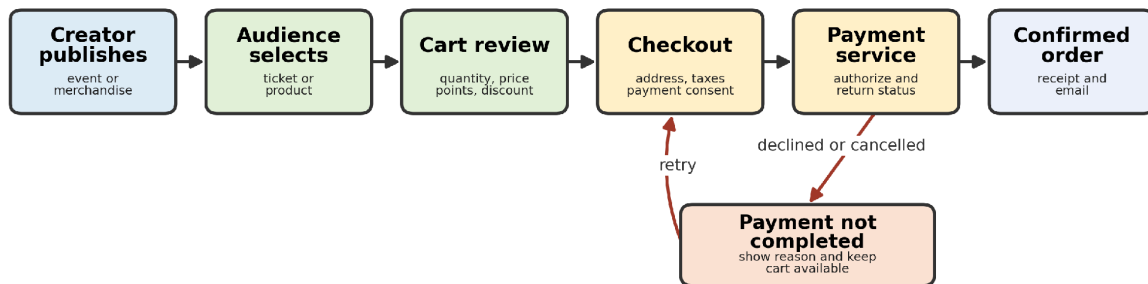
Audience actions for discovery participation commerce connections and loyalty



**Figure 2-4: Audience dashboard workflow**

The audience dashboard combines profile controls, favorite artists, the community feed, connected-app permissions, cart and order history, and a private loyalty balance. Create, view, update, deactivate, and deletion actions remain limited by security, retention, and transaction-record requirements.

## Purchase checkout and confirmation workflow



**Figure 2-5: Purchase checkout and confirmation workflow**

A creator publishes an event ticket or merchandise item, an audience member reviews it in the cart, and checkout sends the payment request to the selected provider. A successful response creates one order, records any loyalty discount, and triggers an in-product receipt and confirmation email. A declined or cancelled payment does not create a fulfilled order and leaves the cart available for correction or retry.

## Administrator dashboard workflow

Managed administrative accounts configure platform rules and resolve exceptions

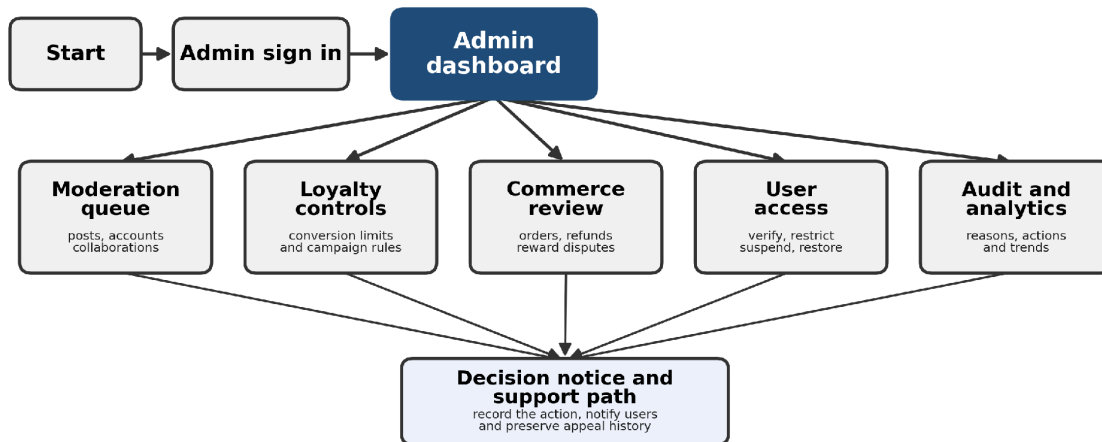


Figure 2-6: Administrator dashboard workflow

The administrator dashboard centralizes moderation, creator verification, loyalty configuration, commerce and reward exceptions, account access, audit history, and basic operational analytics. Every material administrative decision records the actor, time, reason, affected record, user notice, and available support or appeal path.

### 2.4 Operating Environment

The product shall operate as a responsive web application in current desktop and mobile browsers. It uses a secure cloud-hosted application service and database, an email service for verification and order notices, a selected payment service in sandbox or approved live mode, optional identity verification, and optional social, event, or streaming connections after technical and policy review. The live demo shall remain usable when a nonessential connected app is unavailable and shall provide a controlled fallback when checkout or email is temporarily unavailable.

### 2.5 Design and Implementation Constraints

The system shall be designed within the following constraints:

- A two-person hackathon team has limited build and test time.
- External APIs may restrict access, review, commercial use, or behavior tied to rewards.
- The prototype cannot prove every stream, view, share, or attendance event; purchase status is authoritative only when returned by the selected payment flow.
- The product shall not encourage fake engagement or manipulation of another platform.
- Loyalty points use a common platform unit but shall not be described as cash, stored monetary value, or transferable value. Redemption is limited to active discount or benefit rules.

The prototype shall not store full payment card numbers or claim compliance beyond the selected payment provider's approved integration.

- A collaboration brief shall not be presented as legal advice or a replacement for a formal contract.
- Private audience activity and rank shall not be public by default.
- Survey and interview evidence is incomplete, so priorities may change.

## **2.6 Assumptions and Dependencies**

The initial product decisions rely on the assumptions that creators will use another tool when it saves time or improves measurable reach, commerce, loyalty, or collaboration value; audience members will opt in when connected-app permissions, loyalty rules, and checkout terms are understandable; creators can offer discounts without erasing profit; and creators from different disciplines want structured introductions. These assumptions must be tested through the surveys and interviews in Appendix A. Identity, payment, email, and connected-app behavior depends on each provider's current API, permission, review, availability, and commercial-use rules.

### 3 External Interface Requirements

The system connects role-based dashboards to identity, payment, email, and optional external services through a responsive web interface. Every connected source shall identify what information is accessed, why it is needed, how long it is retained, and how the user can disconnect it. Checkout shall clearly identify price, discounts, taxes or fees where known, payment status, and confirmation. The prototype shall use sandbox, simulated, or manually verified data unless a live connection is clearly permitted and tested.

#### 3.1 Data Elements for External Interface

External interface data is organized by purpose rather than by provider so that the system can remain useful when one provider is changed or removed. Optional values may be omitted when they are not required for the chosen feature.

#### High-level application and data exchange structure

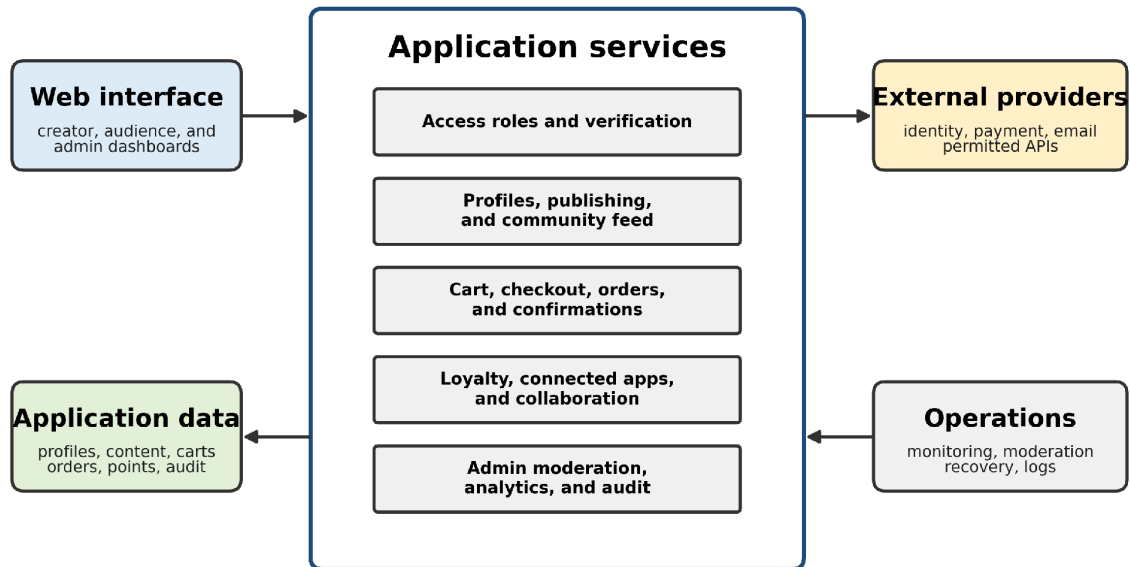


Figure 3-1: High-level application and data exchange structure

Table 3-1: External interface data elements

| Data group      | Required data                                          | Optional data                                                                                                                                                     | Purpose                                        |
|-----------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Account         | Email, role, password hash, account status             | Display name, image                                                                                                                                               | Sign-in and account operation                  |
| Creator profile | Display name, description, creative discipline, skills | Location, time zone, remote or in-person availability, languages, experience level, portfolio links, audience-size range, preferred arrangement, and availability | Identity, discovery, and opportunity filtering |

| Data group          | Required data                                                                                 | Optional data                                                                                                                 | Purpose                                                                                        |
|---------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Published item      | Creator, type, title, status, created time; price and availability for merchandise or tickets | Image, venue, external URL, inventory, rights metadata; for opportunities: arrangement, filters, response limit, and timeline | Posts, events, merchandise, creator-only opportunities, feed display, and checkout eligibility |
| Audience preference | Audience member, creator, favorite state                                                      | Notification choice                                                                                                           | Combined feed and notices                                                                      |
| Activity record     | Source, action, time, status                                                                  | Value, external reference                                                                                                     | Explain reward eligibility and disputes                                                        |
| Reward              | Creator or platform rule, benefit, threshold, quantity, dates                                 | Code, instructions                                                                                                            | Control eligibility and redemption                                                             |
| Safety record       | Reporter, item, reason, status                                                                | Evidence and notes                                                                                                            | Moderation, appeals, and audit                                                                 |
| Connected app       | User, provider, permission scopes, connection status                                          | Provider account reference and last sync                                                                                      | Consent, verification, and disconnect control                                                  |
| Cart                | Audience member, item, quantity, current price                                                | Applied loyalty rule and estimated fees                                                                                       | Review intended purchase before checkout                                                       |
| Order               | Order ID, buyer, creator, items, totals, currency, payment status, created time               | Fulfillment, cancellation, refund, and confirmation references                                                                | Receipt, order history, dispute handling, and creator reporting                                |
| Payment record      | Provider reference, status, amount, currency                                                  | Failure reason category and refund reference                                                                                  | Reconcile checkout without storing full card details                                           |
| Rights metadata     | Creator, item, ownership or credit statement                                                  | License, intended use, territory, and duration                                                                                | Inform collaborators and audiences without claiming legal enforcement                          |

**Table 3-2: Collaboration opportunity and brief data elements**

| Data element              | Description                                                                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Participants              | Every creator involved and the role each person expects to perform                                                                               |
| Goal and deliverables     | What will be produced and how completion will be recognized                                                                                      |
| Timing                    | Target dates, milestones, and availability                                                                                                       |
| Credit                    | How names, handles, links, and contributions will appear                                                                                         |
| Usage                     | Where the work may appear and for how long                                                                                                       |
| Collaboration arrangement | Paid work, skill exchange, revenue sharing, unpaid collaboration, or open to discussion; include proposed payment, expenses, or split when known |
| Approvals                 | Which participant must approve content before release                                                                                            |
| Change history            | Dated record of revisions and confirmations                                                                                                      |

| Data element         | Description                                                                                                                                                  |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eligibility filters  | Profile fields selected as required or preferred, including creative discipline, skills, location, availability, language, experience, or portfolio evidence |
| Response control     | Maximum responses of 5, 10, or 20; default 10; current count; open, paused, closed, or reopened status                                                       |
| Opportunity response | Responding creator, short interest message, linked profile or portfolio, date, and shortlisted or declined status                                            |

## 4 Internal System Features

This section defines the main internal components. Each component includes its purpose, priority, stimulus and response sequence, and testable functional requirements. Requirements beginning with REQ are required for the stated scope. Requirements beginning with OPT are optional or later-version capabilities.

### 4.1 Access Manager

The Access Manager authenticates users, assigns creator, audience, or administrator roles, and prevents unauthorized access to private actions. It also supports recovery, deletion requests, and auditable administrator permissions.

#### 4.1.1 Priority

The Access Manager is high priority because every private feature depends on reliable identity and role checks.

#### 4.1.2 Stimulus Response Sequences

When a person signs up, signs in, resets a password, changes role information, signs out, or requests deletion, the Access Manager validates the request and returns a clear success or error response. When an administrator performs a restricted action, the component confirms administrator permission and records the action.

#### 4.1.3 Functional Requirements

**Table 4-1: Access Manager functional requirements**

| Requirement ID | Description                                                                                                                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-AM-1.1     | The system shall allow a person to create an account with an email address and password.                                                                                                                        |
| REQ-AM-1.2     | The system shall allow a person to choose creator, audience, or both roles. Administrator access shall be provisioned separately.                                                                               |
| REQ-AM-1.3     | The system shall verify the email address before private creator or audience features are available.                                                                                                            |
| REQ-AM-1.4     | The system shall allow a user to reset a forgotten password.                                                                                                                                                    |
| REQ-AM-1.5     | The system shall allow a user to sign out and invalidate the current session.                                                                                                                                   |
| REQ-AM-1.6     | The system shall allow a user to request account deletion.                                                                                                                                                      |
| REQ-AM-1.7     | The system shall restrict administrative actions to authorized administrator accounts.                                                                                                                          |
| OPT-AM-1.8     | The system may allow a user to sign out of all active sessions.                                                                                                                                                 |
| REQ-AM-1.9     | The system shall route a signed-in user to the correct creator, audience, or administrator dashboard and allow a dual-role user to switch between creator and audience views.                                   |
| REQ-AM-1.10    | The system shall require configured creator identity verification before high-risk publishing, loyalty, or payment-related actions and shall expose only verification status, not unnecessary identity details. |

## 4.2 Creator Profile and Publishing

This component lets creators describe their work and publish the items audience members or collaborators need to see. A creator can control whether the profile is visible to audience members, collaborators, or both.

### 4.2.1 Priority

This component is high priority because creator content supplies the main value shown in the audience feed and collaboration search.

### 4.2.2 Stimulus Response Sequences

When a creator creates or edits a profile, announcement, event, offer, or collaboration opportunity, the system validates required fields, saves the draft or published version, and records who changed it and when. Published items become visible only to the intended audience.

### 4.2.3 Functional Requirements

**Table 4-2: Creator Profile and Publishing functional requirements**

| Requirement ID | Description                                                                                                                                                                                                                                                                                                                                    |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-CP-1.1     | The system shall allow a creator to maintain a structured profile containing display name, description, creative discipline, skills, location and time zone, remote or in-person availability, languages, experience level, portfolio links, audience-size range, preferred collaboration arrangements, and availability or expected timeline. |
| REQ-CP-1.2     | The system shall allow a creator to choose whether the profile is public and whether it appears in collaboration search.                                                                                                                                                                                                                       |
| REQ-CP-1.3     | The system shall allow a creator to create, view, edit, publish, unpublish, and delete a community post.                                                                                                                                                                                                                                       |
| REQ-CP-1.4     | The system shall allow a creator to publish an event with date, time, place or online link, price, ticket quantity, and checkout or permitted external ticket link.                                                                                                                                                                            |
| REQ-CP-1.5     | The system shall allow a creator to create, view, edit, publish, unpublish, and delete a merchandise item with images, description, price, inventory status, and fulfillment instructions.                                                                                                                                                     |
| REQ-CP-1.6     | The system shall keep a record of who created or changed a published item and when.                                                                                                                                                                                                                                                            |
| OPT-CP-1.7     | The system may allow a creator to mark an announcement, event, or offer as featured.                                                                                                                                                                                                                                                           |
| REQ-CP-1.8     | The system shall provide creator-dashboard access to profile, publishing, community feed, collaboration, audience insights, loyalty setup, and logout functions.                                                                                                                                                                               |
| REQ-CP-1.9     | The system shall allow a creator to attach and revise rights metadata covering ownership, credit, license, or intended use while clearly stating that the record is informational and not legal enforcement.                                                                                                                                   |
| REQ-CP-1.10    | The system shall allow a creator to view measured post, event, merchandise, order, and audience activity without exposing unnecessary personal or payment information.                                                                                                                                                                         |

## 4.3 Audience Feed and Favorites

This component helps audience members find creators, save favorites, and see a single time-ordered view of important creator updates without depending only on social-media algorithms.

### 4.3.1 Priority

The audience feed is high priority because the product must provide value even when an audience member does not connect any external account.

### 4.3.2 Stimulus Response Sequences

When an audience member searches, favorites a creator, removes a favorite, opens the feed, changes a filter, or reports an item, the system updates the preference and returns only published items the audience is allowed to see.

### 4.3.3 Functional Requirements

**Table 4-3: Audience Feed and Favorites functional requirements**

| Requirement ID | Description                                                                                                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-FF-1.1     | The system shall allow an audience member to search creators by name, discipline, skill, and supplied location.                                                                      |
| REQ-FF-1.2     | The system shall allow an audience member to favorite and unfavorite a creator.                                                                                                      |
| REQ-FF-1.3     | The system shall show published posts, events, and merchandise from favorite creators in a combined community feed.                                                                  |
| REQ-FF-1.4     | The system shall identify the creator, item type, publication time, and relevant external link for every feed item.                                                                  |
| REQ-FF-1.5     | The system shall allow an audience member to report misleading, unsafe, or abusive content.                                                                                          |
| OPT-FF-1.6     | The system may filter the community feed by posts, events, merchandise, and creator-visible collaborations.                                                                          |
| OPT-FF-1.7     | The system may allow notification preferences for each favorite creator.                                                                                                             |
| REQ-FF-1.8     | The system shall provide audience-dashboard access to profile, favorite artists, community feed, connected apps, cart, loyalty points, order history, and logout functions.          |
| REQ-FF-1.9     | The system shall allow an audience member to connect an eligible app with explicit permission scopes, view its status, and disconnect it without deleting unrelated account content. |
| REQ-FF-1.10    | The system shall allow an audience member to view, edit, deactivate, or request deletion of the audience profile, subject to stated order, safety, and audit retention rules.        |

## 4.4 Loyalty and Reward Engine

This component lets creators recognize eligible audience activity and provide creator-approved benefits. It maintains a transparent, private record so an audience member can understand why an activity is pending, approved, rejected, reversed, or redeemed.

#### 4.4.1 Priority

The Loyalty and Reward Engine is high priority because it tests the core idea that recognition and useful benefits can deepen the creator-to-audience relationship. External listening or viewing verification is not required for the prototype.

#### 4.4.2 Stimulus Response Sequences

When a creator defines an eligible activity and reward, the system validates the rule and publishes it. When an audience member completes a directly verifiable, manually verified, or simulated activity, the system records the source and time, applies limits, updates the ledger, and changes eligibility. When the audience redeems, the system prevents a duplicate redemption and records the final state.

#### 4.4.3 Functional Requirements

**Table 4-4: Loyalty and Reward Engine functional requirements**

| Requirement ID | Description                                                                                                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-LR-1.1     | The system shall allow a creator to define one or more eligible audience activities within administrator-configured limits.                                                     |
| REQ-LR-1.2     | The prototype shall support directly verifiable or simulated activities such as RSVP, check-in, purchase code, poll, referral, or mailing-list signup.                          |
| REQ-LR-1.3     | The system shall allow a creator to assign platform loyalty points, status, or a direct benefit to an eligible activity.                                                        |
| REQ-LR-1.4     | The system shall allow a creator to set reward quantity, start date, end date, and eligibility threshold.                                                                       |
| REQ-LR-1.5     | The system shall show each audience member a private ledger containing activity, source, date, point change, discount use, and status.                                          |
| REQ-LR-1.6     | The system shall support pending, approved, rejected, reversed, and redeemed states.                                                                                            |
| REQ-LR-1.7     | The system shall limit repeated activity according to the creator's rule.                                                                                                       |
| REQ-LR-1.8     | The system shall prevent duplicate reward redemption.                                                                                                                           |
| REQ-LR-1.9     | The system shall show a clear reason when an activity is rejected or reversed.                                                                                                  |
| REQ-LR-1.10    | Loyalty points shall use one platform-wide unit, shall not be represented as cash or transferable value, and shall be redeemable only through active discount or benefit rules. |
| OPT-LR-1.11    | The system may allow a creator to pause a reward without deleting historical records.                                                                                           |
| OPT-LR-1.12    | Listening or viewing activity may be added only after provider permission and reliable verification are confirmed.                                                              |
| REQ-LR-1.13    | The system shall allow an authorized administrator to configure permitted point-to-discount conversion ratios, minimums, maximums, expiration rules, and campaign limits.       |
| REQ-LR-1.14    | The cart and checkout shall show the point balance used, discount value, remaining balance, and final total before confirmation.                                                |

| Requirement ID | Description                                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-LR-1.15    | A cancelled, failed, or refunded order shall release or reverse loyalty points according to the active rule and record the reason in the ledger. |

## 4.5 Collaboration Manager

The Collaboration Manager lets creators create an opportunity, join an eligible opportunity, and manage active or previous collaborations. It supports direct creator-to-creator outreach and a creators-only opportunity feed. Opportunity responses are short, project-specific expressions of interest rather than job applications. Audience accounts cannot access private opportunities, responses, conversations, shortlists, or work-in-progress information.

### 4.5.1 Priority

This component is high priority for product differentiation. The MVP supports direct outreach, structured opportunities, profile-based filters, controlled response limits, and a plain-language brief. Automated matching, escrow, payroll, and contract generation remain outside the prototype scope.

### 4.5.2 Stimulus Response Sequences

A creator may contact another creator directly or publish an opportunity to the creators-only feed. The opportunity identifies the project, collaboration arrangement, required and preferred profile filters, and a response limit of 5, 10, or 20, with 10 selected by default. An eligible creator may respond once with a short message and linked profile or portfolio. The response opens a private conversation tied to that opportunity. When the limit is reached, the opportunity pauses automatically until the owner shortlists, declines, closes, or reopens it. A project becomes active only after the selected creators confirm the latest collaboration brief, and every change remains dated and attributable.

### 4.5.3 Functional Requirements

**Table 4-5: Collaboration Manager functional requirements**

| Requirement ID | Description                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-CM-1.1     | The system shall allow a creator to contact another creator directly through a private creator-to-creator conversation.                                                                               |
| REQ-CM-1.2     | The system shall provide a collaboration opportunity feed that is accessible only to creator accounts.                                                                                                |
| REQ-CM-1.3     | The system shall prevent audience accounts from viewing collaboration opportunities, responses, conversations, shortlists, or work-in-progress information.                                           |
| REQ-CM-1.4     | The system shall allow a creator to publish an opportunity with a goal, creative discipline or skills needed, location or remote status, timing, expected deliverable, and collaboration arrangement. |
| REQ-CM-1.5     | The system shall require every opportunity to specify paid work, skill exchange, revenue sharing, unpaid collaboration, or open to discussion.                                                        |
| REQ-CM-1.6     | The system shall allow the opportunity owner to choose a maximum of 5, 10, or 20 responses, with 10 selected by default.                                                                              |

| Requirement ID | Description                                                                                                                                              |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-CM-1.7     | The system shall allow each creator to respond only once to an opportunity using a short interest message and the creator's linked profile or portfolio. |
| REQ-CM-1.8     | The system shall automatically pause an opportunity when its response limit is reached.                                                                  |

**Table 4-5: Collaboration Manager functional requirements (continued)**

| Requirement ID | Description                                                                                                                                                                                                                  |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-CM-1.9     | The system shall allow the opportunity owner to shortlist, decline, close, or reopen the opportunity and its responses.                                                                                                      |
| REQ-CM-1.10    | The system shall allow the opportunity owner to select required and preferred response filters from structured creator-profile fields.                                                                                       |
| REQ-CM-1.11    | The system shall prevent a creator who does not meet a required filter from responding and shall explain which requirement was not met.                                                                                      |
| REQ-CM-1.12    | The system shall use preferred filters to highlight or order stronger matches without blocking otherwise eligible creators.                                                                                                  |
| REQ-CM-1.13    | The system shall create a private conversation linked to the specific opportunity when an eligible creator responds.                                                                                                         |
| REQ-CM-1.14    | The system shall allow the opportunity owner to accept, decline, or request changes from a responding creator.                                                                                                               |
| REQ-CM-1.15    | The system shall record roles, deliverables, dates, approvals, credit, intended usage, and any proposed payment, expenses, skill exchange, or revenue split in a collaboration brief.                                        |
| REQ-CM-1.16    | The system shall require all selected creators to confirm the latest collaboration brief before the project becomes active and shall keep a dated history of changes and approvals.                                          |
| REQ-CM-1.17    | The system shall allow a creator to report a collaborator or withdraw before confirmation and shall state that the collaboration brief is not legal advice or a formal contract.                                             |
| OPT-CM-1.18    | The system may allow all participating creators to approve a completed collaboration for publication as a separate ordinary creator post; the opportunity and its private records shall remain hidden from audience members. |
| REQ-CM-1.19    | The creator dashboard shall separate create collaboration, join collaboration, and manage my collaborations actions and shall show each collaboration's current state.                                                       |

## 4.6 Notifications and Analytics

This component provides essential status notices and simple evidence about creator reach and activity. It avoids unsupported claims that a click caused a sale when the transaction is not directly measured.

### 4.6.1 Priority

Essential account and workflow notifications are high priority. Advanced automation, attribution, and audience segmentation are lower priority.

#### 4.6.2 Stimulus Response Sequences

When a request, reward, security event, or administrator decision changes state, the system creates an in-product notice and an email where appropriate. When a creator opens analytics, the system returns current counts, date range, and data source without exposing unnecessary audience details.

#### 4.6.3 Functional Requirements

**Table 4-6: Notifications and Analytics functional requirements**

| Requirement ID | Description                                                                                                                                                                                                      |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-NA-1.1     | The system shall notify affected users about accepted collaboration requests, loyalty approval or redemption, order confirmation or refund, account security, and administrative action.                         |
| REQ-NA-1.2     | The system shall allow users to turn nonessential notifications on or off.                                                                                                                                       |
| REQ-NA-1.3     | A notification shall link to the relevant item and shall not expose private details in the email subject line.                                                                                                   |
| REQ-NA-1.4     | The system shall show creators measured counts for followers, feed views, event interest, merchandise views, cart additions, confirmed orders, activity completion, and loyalty redemption.                      |
| REQ-NA-1.5     | Analytics shall state the date range and data source.                                                                                                                                                            |
| REQ-NA-1.6     | The system shall not claim a sale was caused by a click unless the sale is directly measured.                                                                                                                    |
| OPT-NA-1.7     | The system may export permissioned contact and campaign data in a common file format after the MVP.                                                                                                              |
| OPT-NA-1.8     | The system may support structured request comments without providing full real-time chat.                                                                                                                        |
| REQ-NA-1.9     | A successful checkout shall create an in-product receipt and send a confirmation email containing the order identifier, items, totals, payment status, and support path without exposing sensitive payment data. |
| REQ-NA-1.10    | If confirmation email delivery fails, the order shall remain available in the user's order history and the system shall record a retryable notification status.                                                  |

## 4.7 Commerce and Order Management

This component manages creator-published tickets and merchandise from audience cart through payment status, order creation, receipt, confirmation email, cancellation, and refund state. Payment credentials remain with the selected provider.

### 4.7.1 Priority

Commerce and order management are high priority because the updated workflow makes the cart, checkout, and confirmation path part of the primary audience experience.

### 4.7.2 Stimulus Response Sequences

When an audience member adds an available ticket or merchandise item, the system records the cart item and shows current quantity, price, loyalty discount, fees where known, and total. At checkout, the system revalidates price and availability, sends the payment request to the selected provider, and creates one order only after a successful response. It then records inventory or ticket changes, an in-product receipt, and confirmation email. Failed or cancelled payment leaves the cart available and does not create a fulfilled order. Refunds and cancellations update the order and loyalty ledger.

### 4.7.3 Functional Requirements

**Table 4-7: Commerce and Order Management functional requirements**

| Requirement ID | Description                                                                                                                                       |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-COM-1.1    | The system shall allow an audience member to add, update, and remove available event tickets or merchandise in a private cart.                    |
| REQ-COM-1.2    | Before checkout, the system shall revalidate item status, quantity, price, loyalty discount, fees where known, and final total.                   |
| REQ-COM-1.3    | The system shall send payment details through the selected provider and shall not store full payment card numbers or security codes.              |
| REQ-COM-1.4    | A successful payment response shall create exactly one order, assign an order identifier, and update applicable ticket or inventory availability. |
| REQ-COM-1.5    | A declined, cancelled, or timed-out payment shall not create a fulfilled order and shall preserve the cart for correction or retry.               |
| REQ-COM-1.6    | The system shall provide an in-product receipt and trigger a confirmation email after successful checkout.                                        |
| REQ-COM-1.7    | The system shall show the audience member order status, items, totals, loyalty discount, payment status, and support path in order history.       |
| REQ-COM-1.8    | The system shall show creators measured order and item counts without exposing full payment credentials or unnecessary audience information.      |
| REQ-COM-1.9    | Cancellation or refund changes shall update the order, inventory where applicable, notification history, and loyalty ledger.                      |

| Requirement ID | Description                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------------------|
| REQ-COM-1.10   | Checkout and provider callbacks shall use idempotency controls so repeated requests cannot create duplicate orders or discounts. |

#### 4.8 Administration and Safety

This component supports user reports, suspicious reward review, account restrictions, appeals, and an audit history. It is intentionally simple enough for a small pilot but must explain significant actions.

##### 4.8.1 Priority

Administration and safety are high priority because creator collaboration and rewards can create fraud, harassment, and dispute risks.

##### 4.8.2 Stimulus Response Sequences

When a user reports content, an account, a reward, or a collaborator, the system adds the item to an administrator queue. An authorized administrator may record a reason and take an allowed action. Significant actions create a notice and support or appeal path.

##### 4.8.3 Functional Requirements

**Table 4-8: Administration and Safety functional requirements**

| Requirement ID | Description                                                                                                                                                              |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-AS-1.1     | The system shall allow administrators to review creator verification, user, content, collaboration, loyalty, order, and reward reports in one role-restricted dashboard. |
| REQ-AS-1.2     | The system shall allow an authorized administrator to warn, restrict, suspend, restore, or remove an account.                                                            |
| REQ-AS-1.3     | The system shall record the administrator, time, reason, and affected item for every administrative action.                                                              |
| REQ-AS-1.4     | The system shall allow a user to block another user from direct interaction.                                                                                             |
| REQ-AS-1.5     | The system shall place suspicious reward activity in pending review before an expensive benefit is issued.                                                               |
| REQ-AS-1.6     | The system shall provide a simple support or appeal path for significant account actions.                                                                                |
| OPT-AS-1.7     | The system may show administrators aggregate platform usage without exposing private audience details unnecessarily.                                                     |
| REQ-AS-1.8     | The system shall allow an authorized administrator to configure allowed loyalty conversion ratios and campaign limits with an effective date and audit history.          |
| REQ-AS-1.9     | The system shall allow an authorized administrator to review order, refund, and loyalty disputes without exposing full payment credentials.                              |
| REQ-AS-1.10    | Every material administrative decision shall record the actor, time, reason, affected record, user notice, and available support or appeal path.                         |

## 5 Nonfunctional Requirements

This section defines qualities that apply across the system. Functional requirements state what the system does; nonfunctional requirements state how safely, reliably, and clearly it must behave.

### 5.1 Performance Requirements

The prototype must remain responsive for the hackathon dataset and reliable during the live demonstration. Performance targets are working acceptance criteria and should be retested with a larger pilot dataset.

**Table 5-1: Performance requirements**

| Requirement ID | Description                                                                                                                       |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------|
| REQ-PER-1.1    | Normal pages shall load within three seconds for most users on a stable connection.                                               |
| REQ-PER-1.2    | Search results shall appear within two seconds for the hackathon dataset.                                                         |
| OPT-PER-1.3    | A confirmed internally processed activity should appear in the loyalty ledger within one minute.                                  |
| REQ-PER-1.4    | The live prototype shall remain usable when an optional external service is unavailable.                                          |
| REQ-PER-1.5    | The system shall prevent duplicate reward redemption during simultaneous requests.                                                |
| REQ-PER-1.6    | After a successful provider response, the order and in-product receipt shall appear within ten seconds for the hackathon dataset. |
| REQ-PER-1.7    | The system shall queue confirmation email for delivery within one minute and preserve the order when email delivery is delayed.   |

### 5.2 Security Requirements

Security requirements protect accounts, private audience activity, creator business information, external service credentials, and administrator actions. The prototype shall use established platform security features rather than inventing custom encryption.

**Table 5-2: Security requirements**

| Requirement ID | Description                                                                                              |
|----------------|----------------------------------------------------------------------------------------------------------|
| REQ-SEC-1.1    | Passwords shall be stored using a recognized one-way password hashing method and never as readable text. |
| REQ-SEC-1.2    | All network traffic shall use encrypted HTTPS connections.                                               |
| REQ-SEC-1.3    | Private actions shall require a signed-in user with the correct role.                                    |
| REQ-SEC-1.4    | Secrets and external service keys shall not be stored in public source code.                             |
| REQ-SEC-1.5    | Sign-in and reward endpoints shall limit repeated automated requests.                                    |
| REQ-SEC-1.6    | Sensitive administrative actions shall be recorded in an audit log.                                      |

| Requirement ID | Description                                                                                                                                                |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-SEC-1.7    | The system shall collect only information needed for a visible account, publishing, connected-app, loyalty, collaboration, cart, order, or safety feature. |
| REQ-SEC-1.8    | Consent shall be separate for each identity, payment, or connected-app purpose and shall identify the requested permission scope.                          |
| REQ-SEC-1.9    | A user shall be able to disconnect a source without losing unrelated account, order, content, or loyalty records.                                          |
| REQ-SEC-1.10   | Private audience activity and rank shall not be public by default.                                                                                         |
| REQ-SEC-1.11   | A deletion request shall remove or de-identify personal information unless a stated legal or safety reason requires limited retention.                     |
| REQ-SEC-1.12   | The system shall use a payment-provider token or hosted flow and shall not store full payment card numbers or security codes.                              |
| REQ-SEC-1.13   | External access tokens and identity-verification references shall be encrypted at rest, access controlled, and excluded from application logs.             |
| REQ-SEC-1.14   | Order, discount, and provider-callback processing shall use integrity and replay protections appropriate to the selected provider.                         |

### 5.3 Usability and Accessibility Requirements

The interface is intended for creators and audience members with different levels of technical experience. Labels and error messages shall use plain language, and essential controls shall be usable without relying on color alone.

**Table 5-3: Usability and accessibility requirements**

| Requirement ID | Description                                                                                                                                                    |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-USA-1.1    | The product shall provide separate but connected creator, audience, and administrator views with role-appropriate navigation.                                  |
| REQ-USA-1.2    | The product shall explain unfamiliar terms when they first appear.                                                                                             |
| REQ-USA-1.3    | A new creator should publish a first item within five minutes during a usability test.                                                                         |
| REQ-USA-1.4    | An audience member should favorite a creator, add an item to the cart, and understand one loyalty discount without instruction.                                |
| REQ-USA-1.5    | Interactive controls shall work with a keyboard and have clear text labels.                                                                                    |
| REQ-USA-1.6    | Text and important controls shall use readable contrast and shall not depend on color alone.                                                                   |
| REQ-USA-1.7    | Error messages shall explain what happened and what the user can do next.                                                                                      |
| REQ-USA-1.8    | The layout shall work on current common mobile and desktop screen sizes.                                                                                       |
| REQ-USA-1.9    | The system shall clearly label simulated data and known prototype limitations.                                                                                 |
| REQ-USA-1.10   | Cart and checkout shall display item quantity, price, applied points, discount, fees where known, and final total before the audience member confirms payment. |

| Requirement ID | Description                                                                                                                                                        |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-USA-1.11   | Connected-app controls shall state the provider, permission purpose, current status, last successful sync when available, and disconnect action in plain language. |

## Appendix A Research Validation

This appendix is intentionally incomplete. Saloni should enter the survey and interview evidence here before the requirements are approved. Blank fields must not be replaced with guesses.

**Table A-1: Survey administration**

| Field                | Creator survey | Audience survey |
|----------------------|----------------|-----------------|
| Launch date          |                |                 |
| Close date           |                |                 |
| Responses            |                |                 |
| Usable responses     |                |                 |
| Recruitment channels |                |                 |
| Sample limitations   |                |                 |

**Table A-2: Survey findings**

| Question or signal                                   | Creator result | Audience result | Requirement or decision change |
|------------------------------------------------------|----------------|-----------------|--------------------------------|
| Difficulty gaining traction                          |                |                 |                                |
| Difficulty reaching engaged followers                |                |                 |                                |
| Interest in cross-genre collaboration                |                | Not applicable  |                                |
| Biggest collaboration blocker                        |                | Not applicable  |                                |
| Interest in one combined creator feed                | Not applicable |                 |                                |
| Preferred audience activities                        |                |                 |                                |
| Preferred rewards                                    |                |                 |                                |
| Comfort connecting an external account               |                |                 |                                |
| Likelihood of trying the product                     |                |                 |                                |
| Prototype volunteers                                 |                |                 |                                |
| Interest in creator merchandise or event checkout    |                |                 |                                |
| Understanding of shared loyalty points and discounts |                |                 |                                |

| Question or signal                     | Creator result | Audience result | Requirement or decision change |
|----------------------------------------|----------------|-----------------|--------------------------------|
| Comfort with connected-app permissions |                |                 |                                |

**A.1 Space for Survey Charts and Observations**

## A.2 Interview Participant Log

Table A-3: Interview log

| ID  | Creator or audience type | Stage or audience size | Date | Main problem | Follow up |
|-----|--------------------------|------------------------|------|--------------|-----------|
| P1  |                          |                        |      |              |           |
| P2  |                          |                        |      |              |           |
| P3  |                          |                        |      |              |           |
| P4  |                          |                        |      |              |           |
| P5  |                          |                        |      |              |           |
| P6  |                          |                        |      |              |           |
| P7  |                          |                        |      |              |           |
| P8  |                          |                        |      |              |           |
| P9  |                          |                        |      |              |           |
| P10 |                          |                        |      |              |           |

## A.3 Interview Findings

| Theme                              | Evidence or quote | Frequency | Product implication |
|------------------------------------|-------------------|-----------|---------------------|
| Audience fragmentation             |                   |           |                     |
| Missed announcements               |                   |           |                     |
| Audience recognition and rewards   |                   |           |                     |
| Cross-genre collaboration          |                   |           |                     |
| Rights, credit, and revenue splits |                   |           |                     |
| Privacy and account connection     |                   |           |                     |
| Willingness to pay                 |                   |           |                     |

## A.4 Validation Decision Log

Table A-4: Validation decision log

| Assumption                                                                       | Evidence | Status      | Decision |
|----------------------------------------------------------------------------------|----------|-------------|----------|
| Creators have a serious audience fragmentation problem                           |          | Unconfirmed |          |
| Creators want cross-genre collaboration support                                  |          | Unconfirmed |          |
| Audience members regularly miss important creator updates                        |          | Unconfirmed |          |
| Audience members will complete valuable activities for creator-approved benefits |          | Unconfirmed |          |
| Audience members will connect an eligible external account                       |          | Unconfirmed |          |
| Creators will fund benefits or pay for the service                               |          | Unconfirmed |          |
| Audience members want a cart and confirmation workflow                           |          | Unconfirmed |          |
| A common loyalty point unit and discount rule is understandable                  |          | Unconfirmed |          |

# Appendix B Survey Instruments

## B.1 Creator Survey

Use the supplied three-minute creator survey. Keep most questions multiple choice or rating scale so the results are easy to graph.

1. Where are you in your creator journey?
2. What kind of content do you create?
3. How difficult is it for you to gain consistent traction?
4. What creates the biggest barrier to your growth?
5. How many separate platforms or tools do you use?
6. How easy is it to identify and directly reach your most engaged followers?
7. How interested are you in collaborating with creators from other genres or specialties?
8. What usually prevents you from collaborating?
9. Which proposed feature would provide the most value: publishing, commerce, audience insights, loyalty, connected apps, or collaboration?
10. How likely are you to try a platform for audience relationships, merchandise or events, loyalty, and collaboration?

Optional follow-up: Would you test an early prototype? Show a contact field only after Yes or Maybe.

## B.2 Audience Survey

Use the supplied three-minute audience survey to test the audience side of the product.

11. How often do you engage with creators online?
12. What kinds of creators do you follow most?
13. How many platforms do you use to follow favorite creators?
14. How often do you miss a release, event, product, or announcement?
15. What prevents you from feeling more connected?
16. How interested are you in one personalized creator feed?
17. Which actions would you complete to earn platform loyalty points or creator-approved benefits?
18. Which rewards would motivate you most, including discounts at checkout?
19. How comfortable are you connecting an eligible app when the requested permissions and disconnect option are clearly explained?

How valuable would a single cart and order history for creator events and merchandise be?

20. How likely are you to try the proposed app?

Optional follow-up: Would you test an early prototype? Show a contact field only after Yes or Maybe.

## Appendix C Traceability and References

**Table C-1: Requirement traceability**

| Product goal                            | Requirement groups              | Research evidence still needed                                                                               |
|-----------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------|
| Direct creator-to-audience relationship | CP, FF, NA, SEC                 | How often creators lose reach and whether audience members want a combined feed and favorite-artist view     |
| Meaningful loyalty                      | LR, COM, NA, AS, SEC            | Which activities create value, whether shared points are understood, and which checkout discounts are useful |
| Cross-genre collaboration               | CM, AM, AS                      | Real projects, trust needs, payment expectations, and rights concerns                                        |
| Creator business value                  | CP, COM, NA, external interface | Current commerce tools, purchase friction, order-confirmation needs, and willingness to pay                  |
| Trust and safety                        | AM, AS, SEC                     | Comfort with data connection and expected support process                                                    |
| Hackathon feasibility                   | All REQ requirements            | Technical estimate and two usability rehearsals                                                              |
| Role-based experience                   | AM, CP, FF, AS, USA             | Whether creator, audience, and administrator dashboards match expected tasks and terminology                 |
| Creator commerce                        | CP, COM, LR, NA, SEC            | Interest in merchandise or event checkout, discount behavior, payment trust, and support expectations        |

### C.1 Working Product Decisions

| Decision                   | Current recommendation                                                                                      | Owner           | When to confirm                   |
|----------------------------|-------------------------------------------------------------------------------------------------------------|-----------------|-----------------------------------|
| Product name               | Use Project Placeholder in technical documents                                                              | Saloni          | Before pitch deck lock            |
| Primary beachhead          | Event-led independent creators, with cross-genre support                                                    | Saloni          | After first ten creator responses |
| Collaboration scope        | Lightweight MVP workflow; full marketplace later                                                            | Saloni and Arya | After research review             |
| First reward action        | Use one platform loyalty point unit with administrator-controlled discount conversion and creator campaigns | Saloni          | After survey                      |
| First external integration | Use sandbox identity, payment, email, and connected-app data unless live permission is clearly confirmed    | Arya            | Before build freeze               |
| Business model             | Free pilot, then subscription, campaign fee, or transaction fee for native measurable sales                 | Saloni          | After interviews                  |

| Decision              | Current recommendation                                                                                | Owner           | When to confirm       |
|-----------------------|-------------------------------------------------------------------------------------------------------|-----------------|-----------------------|
| Checkout approach     | Use a selected payment-provider sandbox first; create orders only after successful provider status    | Arya            | Before build freeze   |
| Confirmation approach | Provide in-product order history and email confirmation; do not let email failure invalidate an order | Saloni and Arya | Before demo rehearsal |

## C.2 References

- IEEE Software Requirements Specification structure and visual format derived from the user-supplied SRS.pdf. Transportation-specific example content was not carried into this product specification.
- User-supplied creator and audience research notes, including the shortened creator and audience surveys.
- Creator Economy Platform Market Research 2026 prepared for this project, covering creator memberships, commerce, communities, audience tools, live events, music products, and collaboration marketplaces.

Updated workflow diagrams supplied September 26, 2026, covering creator, audience, administrator, loyalty, commerce, and confirmation paths.

- Current external-platform permissions, privacy rules, and commercial-use terms must be checked again immediately before any live integration is implemented.